

STAINLESS STEEL MANUFACTURING · STEEL MELTING SHOP

# AOD Operations

Process Analysis, Standardisation and Business Requirements

**Portfolio edition.** This is a sanitised version of work carried out in an operations role at a large stainless steel producer. The employer is not named, and all process parameters, quantities, temperatures, gas ranges, equipment identifiers and timings shown are illustrative placeholders, not operating values. The analysis structure, business rules, requirements, controls and validation logic reflect the work as performed. Nothing commercially sensitive is reproduced.

Shiv Satyam Singh  
Associate Manager

# About This Document

This document consolidates three pieces of process and requirements work carried out in the AOD (Argon Oxygen Decarburisation) area of a Steel Melting Shop. Each part addresses a distinct operational problem, and each follows the same structure: business problem, objectives and scope, stakeholders, current-state analysis, future-state design, rules and controls, and the measures used to confirm the change held.

The AOD converter is the point at which carbon is removed from the melt and the final chemistry of the heat is set. Material additions, gas ratios and temperature windows differ by grade family, and the process runs continuously across three shifts. The recurring problem across all three pieces of work is the same one: operating knowledge sat with experienced individuals rather than in a controlled reference, so execution varied by shift and by operator.

The work was performed in an operations role, using structured requirements, business rules, traceability and acceptance criteria to document the process and support consistent execution across shifts.

## Document Control

| Field            | Detail                                                                                                                                      |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared by      | S. Satyam Singh                                                                                                                             |
| Role at the time | Associate Manager, AOD Operations                                                                                                           |
| Business area    | Steel Melting Shop, AOD Operations                                                                                                          |
| Coverage         | Grade-wise addition pattern standardisation; AOD auto batching requirements; 5S material allocation                                         |
| Methods applied  | Process mapping (As-Is / To-Be), requirements definition, business rules, DMAIC, RACI, control planning, validation scenarios, traceability |
| Grade families   | 200 series, 300 series, 400 series                                                                                                          |
| Version          | 2.0 (portfolio edition)                                                                                                                     |

## Contents

| Part | Subject                         | Focus                                                                                                                                               |
|------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Grade-Wise AOD Addition Pattern | Process mapping and standardisation of material addition sequence, temperature control and gas switch-over across three grade families              |
| B    | AOD Auto Batching               | Business and functional requirements, bunker and weighing-hopper logic, exception handling and validation scenarios for automated material batching |
| C    | 5S Material Allocation at AOD   | Current-state analysis, gap analysis and requirements for material location control, receipt handover and space utilisation                         |

## Terms Used

| Term                   | Meaning                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| AOD                    | Argon Oxygen Decarburisation, the converter process used to refine stainless steel     |
| Main blow / DB1 / DB2  | Primary oxygen blowing stage, followed by second and third blow stages                 |
| Zero batch             | Initial flux charge added before pouring                                               |
| Dololime / Lime / Flux | Slag-forming additions used to control chemistry and protect refractory                |
| Ferroalloys            | Chromium, nickel and manganese bearing additions used to achieve target chemistry      |
| MHS                    | Material Handling System: conveyors, bunkers, feeders and chutes serving the converter |
| CH1 / CH2              | Charging hoppers feeding material into the converter                                   |
| Weighing hopper        | Intermediate hopper that weighs material before transfer to the charging chute         |
| Reduction              | Post-blow stage where chromium is recovered from slag                                  |

PART B

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AOD Auto Batching

Business and functional requirements, operating rules, exception handling and validation scenarios for automated material batching during the main blow window.

## B1. Background

AOD operations require metallic and flux additions during the main blow period. Automated batching depends on accurate material selection, correct bunker configuration, appropriate feed and discharge rates, and clear trigger points tied to the oxygen phase.

For the sequence to run without interruption, material shifting, charging hopper movement, feeder condition, belt status, flame protection gates and auto-mode readiness must all be controlled before and during the process.

## B2. Business Problem

The operational challenge was to control auto batching steps repeatably so that material additions complete within the expected main blow window. Without a structured process view, teams faced ambiguity around bunker priority, weighing hopper usage, colour status interpretation, feeder trips, charging hopper readiness and step table reconciliation.

| Problem area                     | Business impact                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Material addition timing         | Delay in completing main blow additions and risk of process variation                    |
| Bunker and weighing hopper logic | Incorrect priority or simultaneous charging created operational confusion                |
| Status visibility                | Operators required clear colour-coded status to understand the material movement stage   |
| Charging hopper control          | Material must be fully discharged and tare weight zero before the next shift of material |
| Safety and exception handling    | Feeder trip, converter tilting and power failure required clear control actions          |
| Evidence and reconciliation      | Step table output required tallying against the discharge table after addition           |

## B3. Objectives

| Objective                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------|
| Document a standard process for AOD auto batching execution                                                     |
| Define business rules for bunker priority, material shifting, weighing hopper use and charging hopper readiness |
| Improve visibility by documenting colour-coded process status rules                                             |
| Support completion of metallic addition and major flux addition within the main blow period                     |
| Reduce operational ambiguity between AOD, MHS, Electrical, Mechanical and shift teams                           |
| Create validation scenarios usable for process checks, training and system acceptance                           |

## B4. Scope

| In scope                                                              | Out of scope                                                     |
|-----------------------------------------------------------------------|------------------------------------------------------------------|
| Auto batching process flow for all three grade families               | Steel grade chemistry design and metallurgical formula ownership |
| Feed rate, discharge rate and material quantity requirements          | Procurement and supplier commercial decisions                    |
| Bunker, weighing hopper and charging hopper operating rules           | PLC coding and automation software development                   |
| Colour-coded status interpretation                                    | Hardware design of conveyors, gates and feeders                  |
| Exceptions including feeder trip, converter tilting and power failure | Final approval authority for plant operating standards           |
| Validation scenarios for process acceptance                           | Financial business case beyond process improvement logic         |

## B5. Stakeholders

| Stakeholder            | Role in process                                          | Requirement focus                                                  |
|------------------------|----------------------------------------------------------|--------------------------------------------------------------------|
| Shift-In-Charge        | Owns shift-level execution and decision making           | Clear steps, exceptions, tally checks and safety controls          |
| AOD Operator           | Runs material addition steps and monitors the converter  | Oxygen trigger points, charging hopper readiness, discharge status |
| MHS Team               | Responsible for material handling equipment and feeders  | Feeder health, belt status, bunker movement, chute availability    |
| Electrical Team        | Supports electrical readiness and flame protection gates | Auto mode, gate health, power-related exception response           |
| Mechanical Team        | Supports converter positioning and mechanical issues     | Cooling position, back pressure checks, emergency handling         |
| Production Management  | Tracks productivity and process performance              | Main blow timing, process discipline, shift compliance             |
| Quality / Process Team | Checks grade requirements and output stability           | Controlled chemistry and consistency of material addition          |

## B6. Step Table Structure

Structure as documented. All values below are illustrative placeholders and do not represent operating parameters.

| Step  | Bunker                                                                       | Trigger                | Material          | Qty | Feed rate | Discharge rate | Shift time | Remarks                                                   |
|-------|------------------------------------------------------------------------------|------------------------|-------------------|-----|-----------|----------------|------------|-----------------------------------------------------------|
| S0    | B-a                                                                          | Prior heat, third blow | Slag former       | Q1  | n/a       | n/a            | 0          | Shift during previous heat                                |
| S1    | B-b                                                                          | Early oxygen band      | Slag former       | Q1  | n/a       | n/a            | 0          | Shift when pouring starts; hold reserve on charging chute |
| S2    | B-c & B-d                                                                    | Oxygen band 2          | Chromium alloy    | Q2  | F1        | D1             | T1         | Shift as soon as confirmed                                |
| S3    | B-e & B-f                                                                    | Oxygen band 3          | Metallic addition | Q3  | F2        | D2             | T2         |                                                           |
| S4    | B-g & B-h                                                                    | Oxygen band 4          | Manganese alloy   | Q4  | F1        | D1             | T3         |                                                           |
| S5    | B-i                                                                          | Oxygen band 5          | Flux              | Q5  | F3        | D3             | T4         |                                                           |
| Total | Expected total process time reference, compared against the main blow window |                        |                   |     |           |                |            |                                                           |

A separate step table was maintained for each of the three grade families. The 300 series table carries additional nickel-bearing steps and therefore a longer expected total; the 400 series carries the shortest.

## B7. As-Is and To-Be Process View

| Area                 | As-Is observation                                                                       | To-Be requirement                                                                     |
|----------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Process reference    | Operational rules held across notes, step tables and team knowledge                     | A single structured process document for execution and training                       |
| Material movement    | Material shifting depended on correct understanding of bunker and weighing hopper rules | Defined priority rules and conditions for simultaneous or sequential bunker operation |
| Status understanding | Operators interpreted colour-coded states from experience                               | Each status colour and its meaning documented                                         |
| Safety controls      | Critical exceptions handled through SOP and shift experience                            | Exception rules included in the process control section                               |
| Validation           | Step table and discharge table comparison required after addition                       | Reconciliation defined as a formal post-addition control                              |

## B8. Functional Requirements

| ID    | Area                    | Requirement                                                                                                       | Priority |
|-------|-------------------------|-------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | Series selection        | The process shall support auto batching step tables for each grade family                                         | High     |
| FR-02 | Material setup          | The process shall define material name, quantity, bunker, feed rate and discharge rate for each step              | High     |
| FR-03 | Oxygen trigger          | Each material addition step shall be linked to a defined oxygen trigger or process event                          | High     |
| FR-04 | Colour status           | The process shall follow colour-coded status for filling, shifting, ready or discharging, and completed states    | High     |
| FR-05 | Bunker priority         | Where multiple bunkers use the same weighing hopper in one step, a fixed left-to-right priority shall be followed | High     |
| FR-06 | Parallel bunker use     | Where bunkers use different weighing hoppers in one step, simultaneous start shall be allowed                     | Medium   |
| FR-07 | Charging hopper control | Material shall shift only after the previous material is fully discharged and tare weight is zero                 | High     |
| FR-08 | Belt status             | Material shall shift only when belt status is green; red status shall block shifting                              | High     |
| FR-09 | Auto mode               | Equipment required for auto batching shall remain in auto mode unless exception handling is active                | High     |
| FR-10 | Post-addition check     | After addition, the step table shall be tallied against the discharge table                                       | High     |
| FR-11 | Tilting control         | During converter tilting in main blow, both charging hoppers shall be discharge-stopped before tilting            | High     |
| FR-12 | Emergency handling      | During power failure or critical interruption, additions shall stop and emergency checks shall be initiated       | High     |

B9. Business Rules

| Rule ID | Business rule                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------|
| BR-B01  | Metallic addition and major flux addition must be completed within the main blow planning window               |
| BR-B02  | A dedicated bunker is allocated to the primary slag former as part of the auto batching configuration          |
| BR-B03  | Bunkers not required for a specific grade family must be marked out of service                                 |
| BR-B04  | Status colour 1 means weighing hopper filling is in progress or not yet complete                               |
| BR-B05  | Status colour 2 means material is shifting to the charging chute                                               |
| BR-B06  | Status colour 3 means all material has shifted to a charging hopper                                            |
| BR-B07  | Status colour 4 means material is ready to discharge or is discharging                                         |
| BR-B08  | Status colour 5 means the step is complete                                                                     |
| BR-B09  | For the same weighing hopper, bunker priority follows a fixed sequence and must not start simultaneously       |
| BR-B10  | For different weighing hoppers, bunkers may start simultaneously within the same step                          |
| BR-B11  | The feeder must not trip during charging or discharging; MHS owns feeder readiness                             |
| BR-B12  | The outlet flame protection gate must be healthy and inlet gates manually open where required                  |
| BR-B13  | After zero batch is shifted to a charging hopper, the empty source must be marked out of service               |
| BR-B14  | On main blow power failure, the lance emergency control must be operated so the lance returns to rest position |

B10. Data Requirements

| Data field          | Purpose                                              |
|---------------------|------------------------------------------------------|
| Series              | Identifies the applicable grade family step table    |
| Step                | Defines sequence number for the batching step        |
| Bunker              | Identifies source bunker for material                |
| Oxygen trigger      | Defines process event or oxygen trigger for addition |
| Material name       | Defines material to be added                         |
| Material quantity   | Defines target material quantity                     |
| Feed rate           | Defines material feeding rate                        |
| Discharge rate      | Defines belt discharge rate                          |
| Time to shift chute | Calculates expected shift time                       |
| Remark              | Captures execution condition or operational note     |

B11. Non-Functional Requirements

| ID     | Category        | Requirement                                                                                                        |
|--------|-----------------|--------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Safety          | The process must prevent unsafe material shifting during converter tilting, power failure or equipment abnormality |
| NFR-02 | Usability       | Status colours and step tables must be readily interpretable by operators and Shift-In-Charges                     |
| NFR-03 | Traceability    | Step table and discharge table must be reconcilable after addition                                                 |
| NFR-04 | Reliability     | Feeder, gates and auto-mode readiness must be verified to reduce process interruption                              |
| NFR-05 | Maintainability | Business rules must be straightforward to update when grade, bunker or material configuration changes              |

B12. Exception Handling and Control Plan

| Exception                                  | Immediate control                                                                                                               | Owner                                     | Evidence                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------|
| Feeder trip during charging or discharging | Stop, check feeder health and resolve before continuing material movement                                                       | MHS Team                                  | Equipment status and shift log |
| Belt shows red status                      | Do not shift material until status is green                                                                                     | AOD Operator / MHS                        | Status indication              |
| Material not fully discharged              | Do not shift next material until discharge is complete and tare weight is zero                                                  | AOD Operator                              | Tare weight check              |
| Converter tilting during main blow         | Both charging hoppers must be discharge-stopped before tilting                                                                  | Shift-In-Charge / AOD Operator            | Operator confirmation          |
| Power failure during blowing               | Stop all additions, confirm emergency gas valve, call Electrical and Mechanical, move converter to cooling position as required | Shift-In-Charge / Electrical / Mechanical | Power failure response log     |
| Main blow power failure                    | Operate the lance emergency control so the lance returns to rest position                                                       | AOD Operator                              | Emergency action confirmation  |

B13. Validation Scenarios

| ID   | Precondition                                  | Action                                     | Expected result                                                                          |
|------|-----------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------|
| V-01 | A grade family is selected                    | Run that family's step table               | Sequence follows S0 to the final step and total time falls within the expected reference |
| V-02 | 300 series selected                           | Run the 300 series step table              | Nickel, chromium, metallic, manganese and flux steps follow the defined sequence         |
| V-03 | Same weighing hopper used by multiple bunkers | Trigger all three bunkers in the same step | The first starts, then the others in sequence; simultaneous start is blocked             |
| V-04 | Different weighing hoppers used               | Trigger all three bunkers in the same step | All permitted bunkers may start simultaneously                                           |
| V-05 | Belt status red                               | Attempt material shift                     | Material shift is not allowed                                                            |
| V-06 | A charging hopper holds a material balance    | Attempt next material shift                | Process waits until material is fully discharged and tare weight is zero                 |
| V-07 | Converter tilting during main blow            | Initiate tilting control                   | Both charging hoppers are discharge-stopped before tilting                               |
| V-08 | Power failure during blowing                  | Apply emergency process                    | Additions stop, emergency gas valve is checked, Electrical and Mechanical are called     |
| V-09 | Addition complete                             | Compare step table with discharge table    | Material quantities and steps reconcile                                                  |

B14. Requirements Traceability

| Requirement | Linked rule or control                      | Validation | Business value                                      |
|-------------|---------------------------------------------|------------|-----------------------------------------------------|
| FR-01       | Series-wise step tables                     | V-01, V-02 | Supports grade-wise process control                 |
| FR-02       | Material setup table                        | V-01, V-02 | Ensures quantity and material detail are controlled |
| FR-04       | Colour status rules                         | V-05       | Improves operator visibility                        |
| FR-05       | Same weighing hopper priority rule          | V-03       | Controls sequential bunker operation                |
| FR-06       | Different weighing hopper simultaneous rule | V-04       | Controls parallel bunker operation                  |
| FR-07       | Tare weight and discharge completion        | V-06       | Prevents incorrect material movement                |
| FR-10       | Step table against discharge table tally    | V-09       | Creates post-addition evidence                      |
| FR-11       | Tilting control                             | V-07       | Controls safety risk                                |
| FR-12       | Emergency handling                          | V-08       | Controls power failure response                     |

B15. RACI Matrix

| Activity                                  | Shift-In-Charge | AOD Operator | MHS | Electrical | Mechanical | Production |
|-------------------------------------------|-----------------|--------------|-----|------------|------------|------------|
| Confirm grade and series step table       | A               | R            | C   | I          | I          | C          |
| Verify material and bunker readiness      | A               | R            | R   | I          | I          | C          |
| Check auto mode and flame gates           | A               | R            | C   | R          | C          | I          |
| Run material shifting and discharge steps | A               | R            | R   | I          | I          | I          |
| Handle feeder trip                        | A               | C            | R   | C          | C          | I          |
| Handle power failure                      | A               | R            | C   | R          | R          | I          |
| Tally step table and discharge table      | A               | R            | C   | I          | I          | C          |

R = Responsible · A = Accountable · C = Consulted · I = Informed